

OUTDOOR ROOMBA (RUMBLE)

TEAM ID: SDDEC25-16

Advisor's name: Dr. Diane Rover





Sudipta Halder: CPRE, Simulation Developer



Brodie Bates: CPRE, *Embedded Engineer, Designer*



Mitchell Owen: EE, Tester, *Designer*



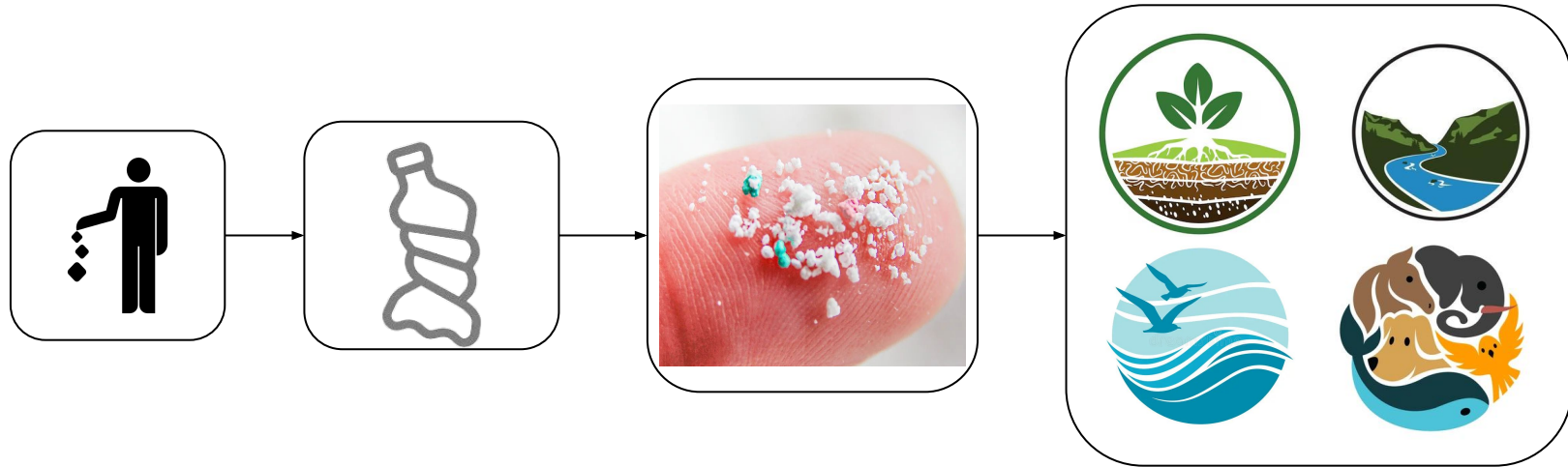
**Daniel Vergara Pinilla: CPRE, *Machine Learning Engineer.*
Back-end Engineer**



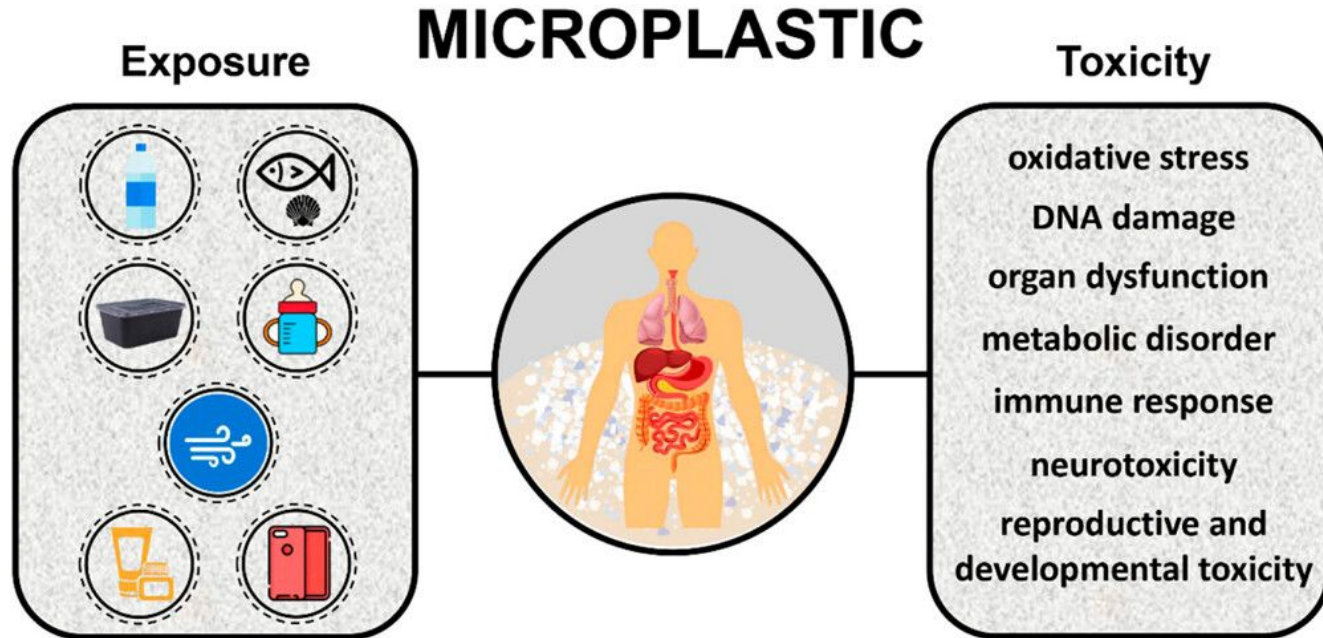
Om Shukla: SE Front-end Engineer

Project Overview

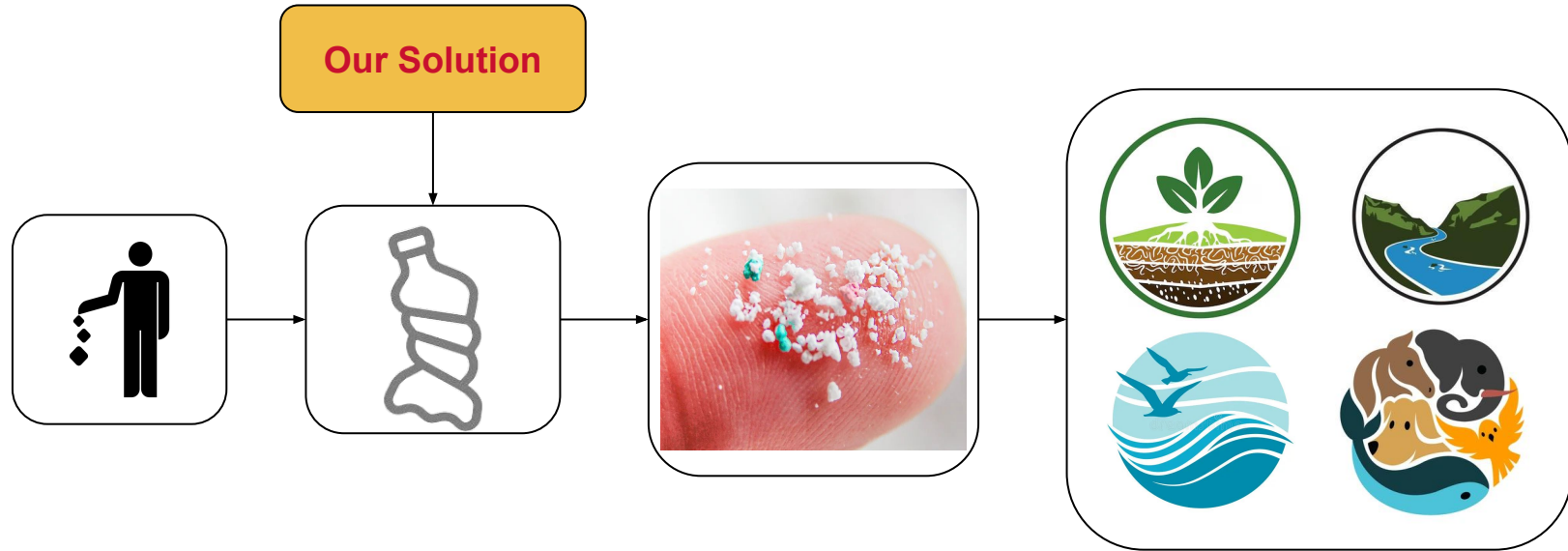
PROBLEM STATEMENT



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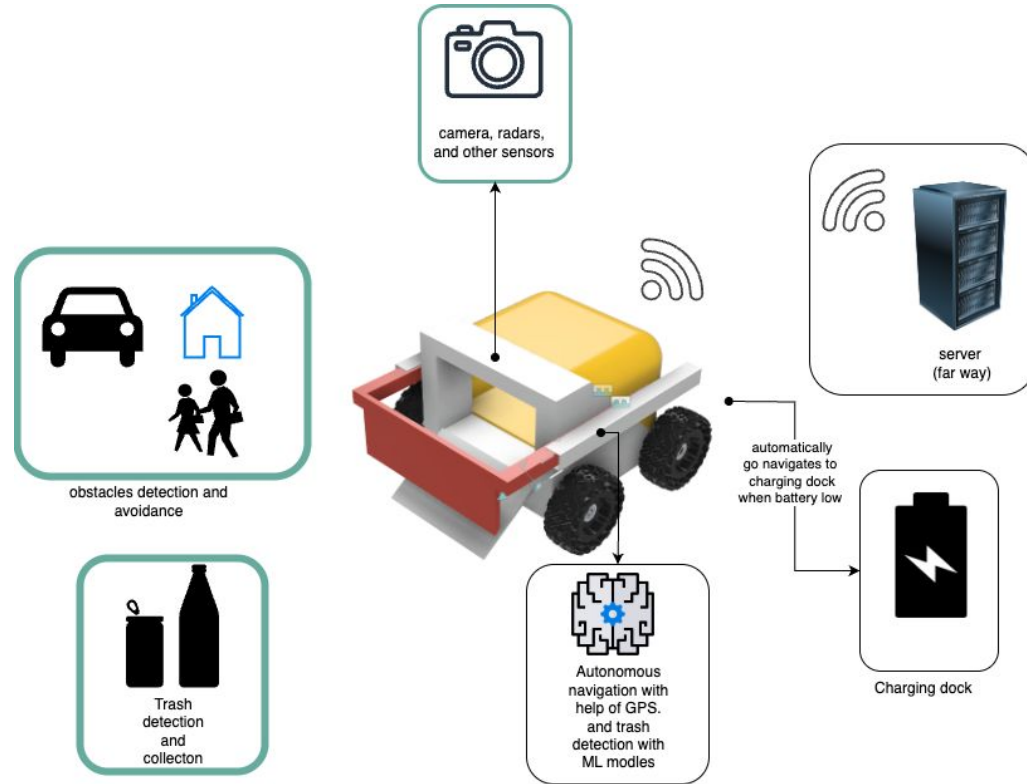


PROPOSED SOLUTION

Create a Robot that picks up trash

Rumble: Outdoor Roomba

- Autonomous
- Object and Trash Recognition
- Refuse Collector
- Unified Endpoint Management
- Self Charging



USERS

- Bars and Restaurants
 - AJ
 - Sips & Paddys
 - BNC
- Event Managers
 - Iowa State
 - Campustown
- Current User
 - Superdog



REQUIREMENTS-HARDWARE

- Object Detection
 - Obstacles
 - Trash
- Power Management
 - 3 hr Battery Life
 - Charging Port
- Durability
 - IPx4



Protected against
water splashing from
any angle.

4

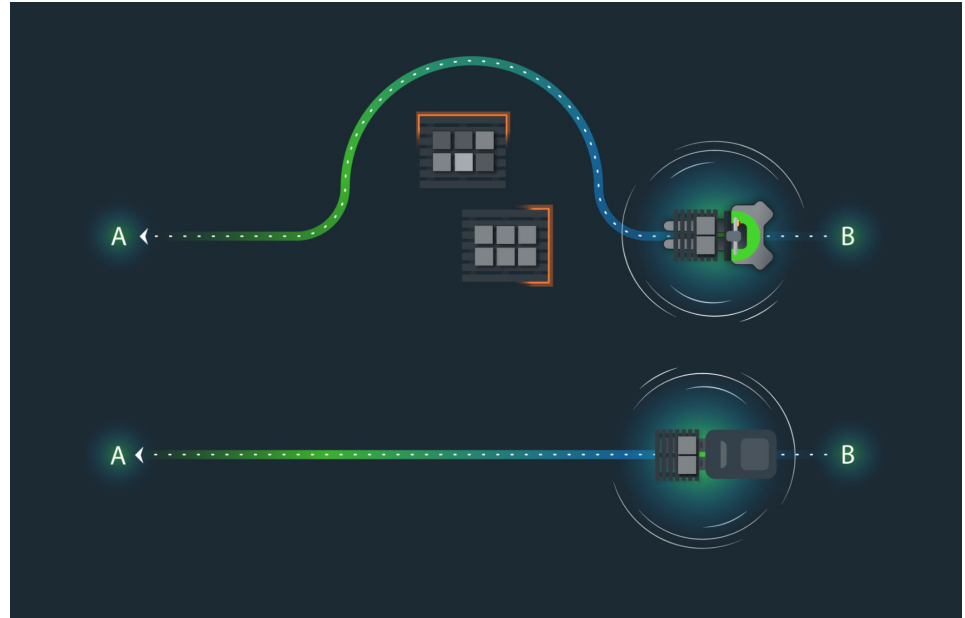
IP Rating Chart

IP ratings are represented by combining the first and second digits of the following columns. See example below.

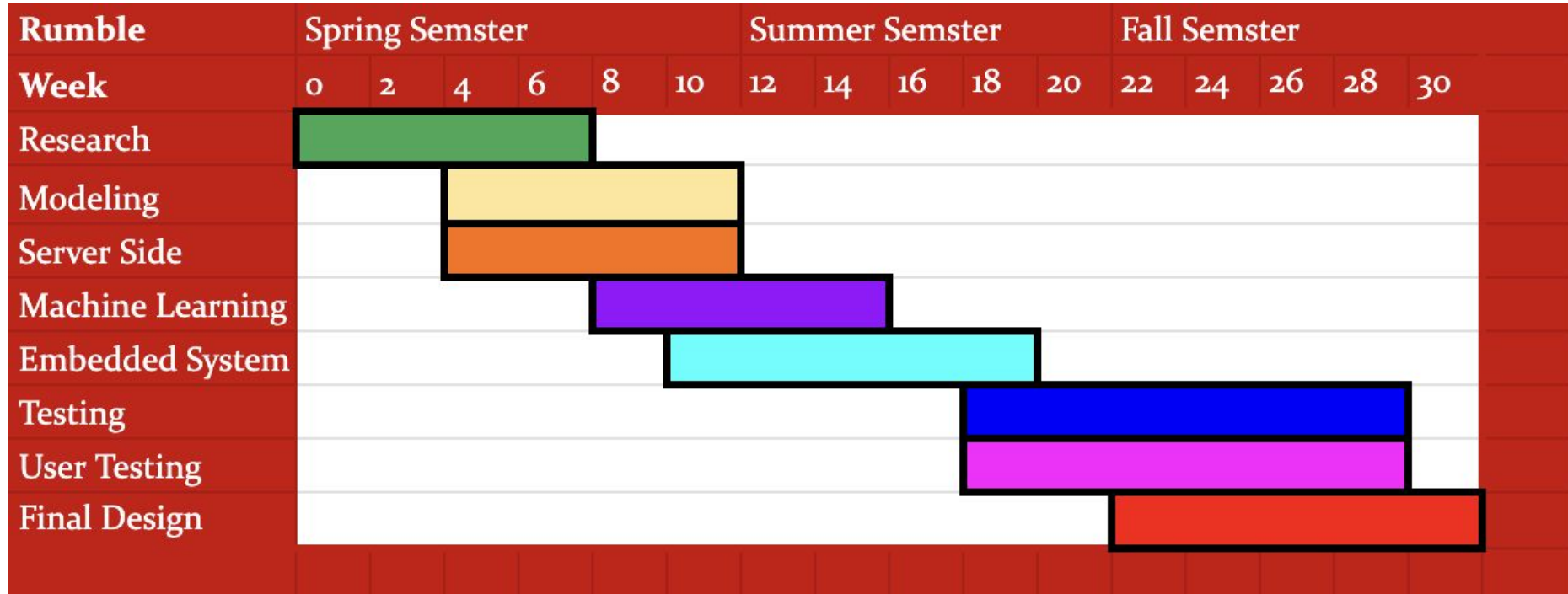
1st Digit - SOLID Degree of protection against solid objects	2nd Digit - LIQUID Degree of protection against water
 No Protection 0	 No Protection 0
 Protected against a solid object greater than 50mm, such as a hand. 1	 Protected against water drops. 1
 Protected against a solid object greater than 12.5mm, such as a finger. 2	 Protected against water drops at a 15 degree angle. 2
 Protected against a solid object greater than 2.5mm, such as a wire. 3	 Protected against water spray at 60 degree angle. 3
 Protected against a solid object greater than 1.0mm, such as a thin strap. 4	 Protected against water splashing from any angle. 4
 Dust Protected. Prevents ingress of dust sufficient to cause harm. 5	 Protected against water jets from any angle. 5
 Dust tight. No ingress of dust. 6	 Protected against powerful water jets and heavy seas. 6
Example:  IP 6 5 Protected against water jets from any angle. Dust tight. No ingress of dust.	
	 Protected against the effects of temporary submersion in water. (30 minutes at 3 feet) 7
	 Protected against the effects of permanent submersion in water. (Up to 13 feet) 8

REQUIREMENTS-SOFTWARE

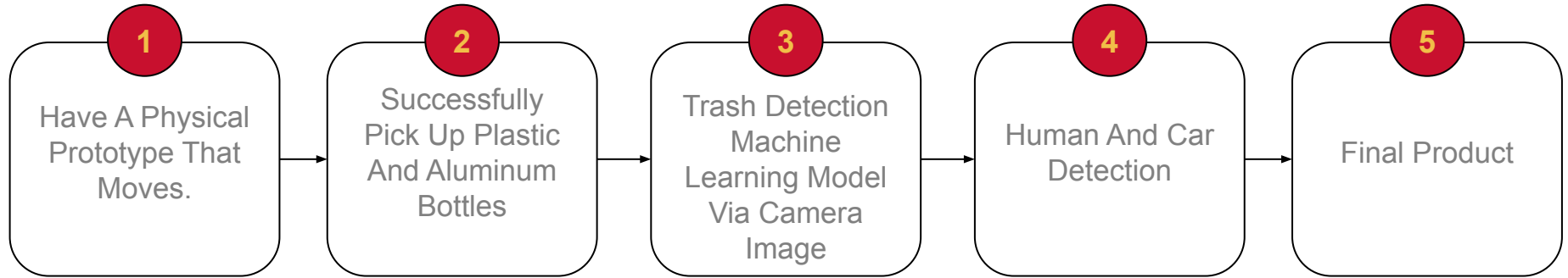
- GPS Tracking
- Object Identification
- Statistic Tracking
- Collisionless Navigation



Project Plan



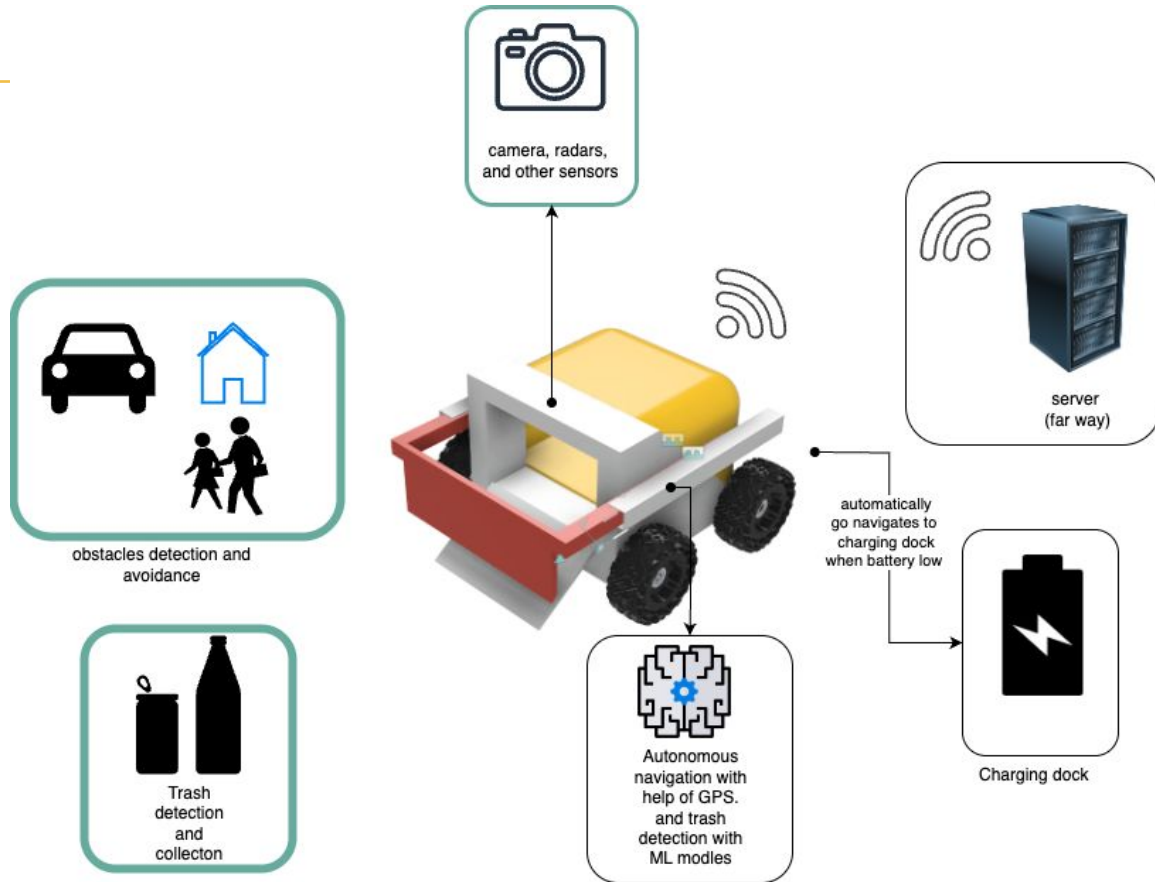
MILESTONES



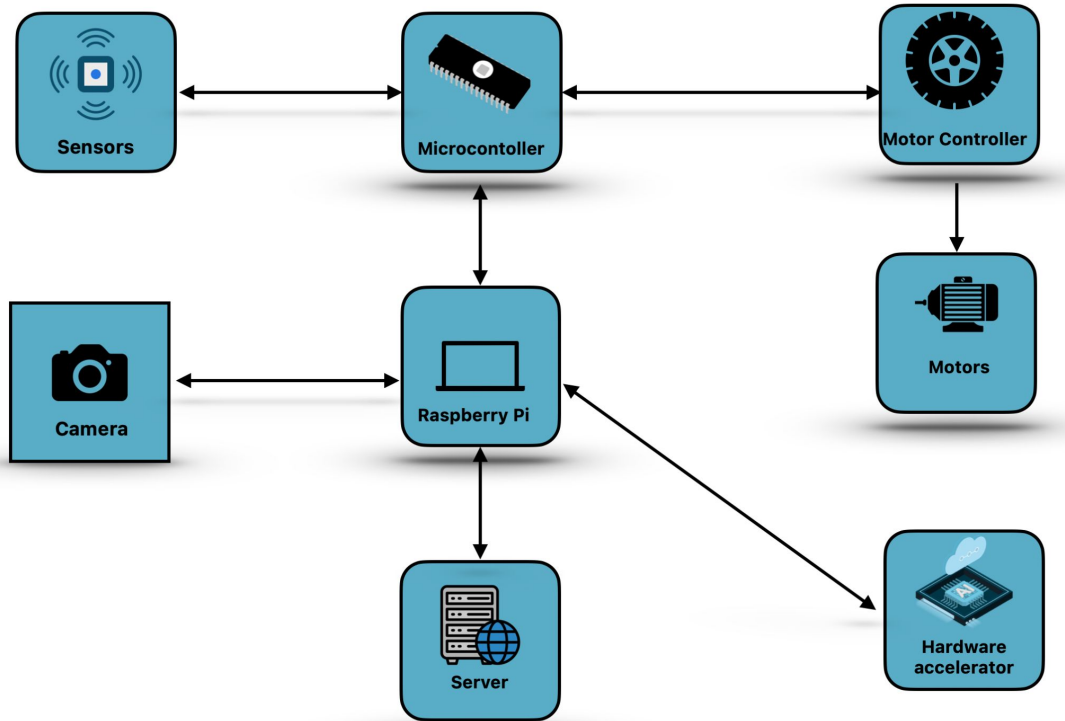
Design

DESIGN OVERVIEW

- System Diagram
- Mechanism
- Machine Learning
- Backend
- Frontend

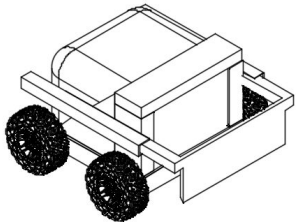
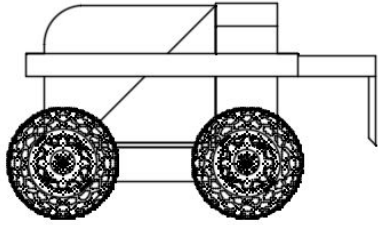


SYSTEM DIAGRAM

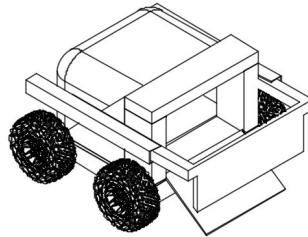
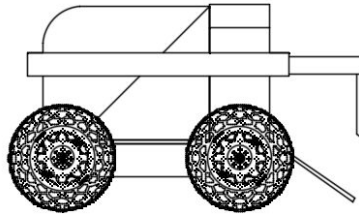


MECHANISM

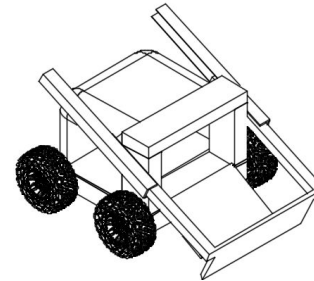
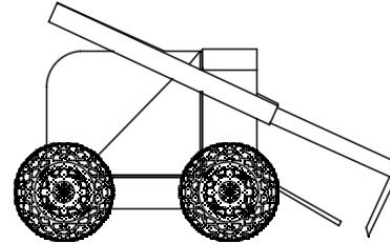
1. Start



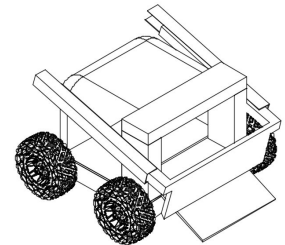
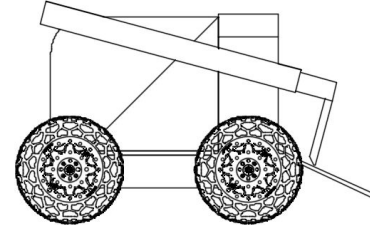
2. Drop Ramp



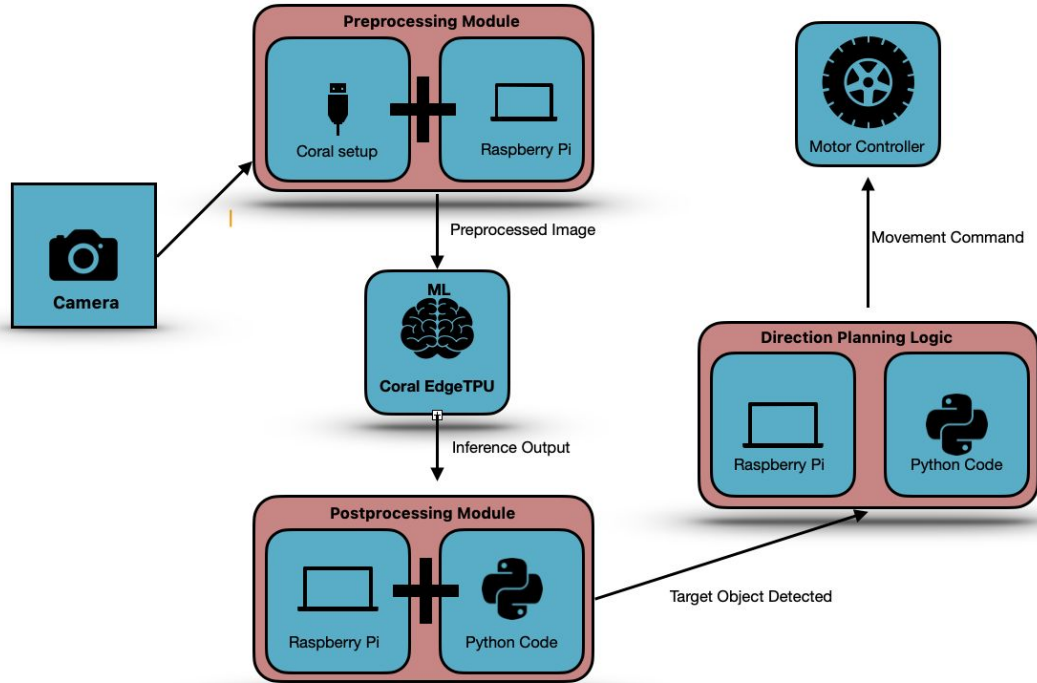
3. Extend Sweep



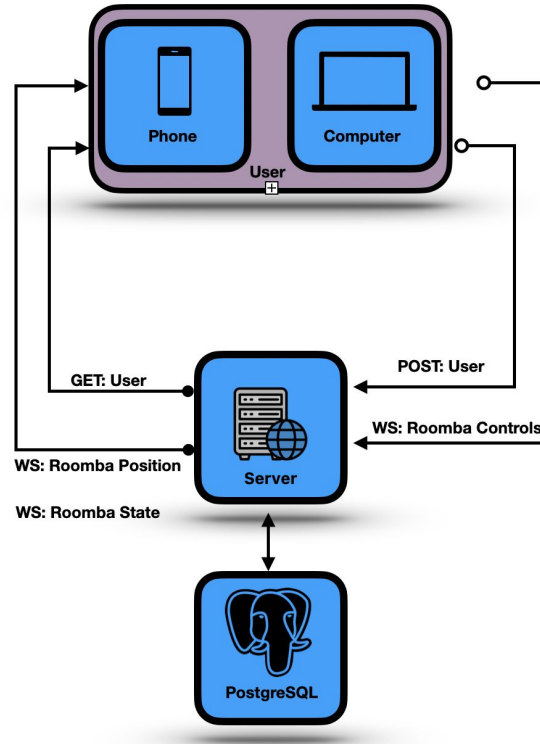
4. Pull Sweep



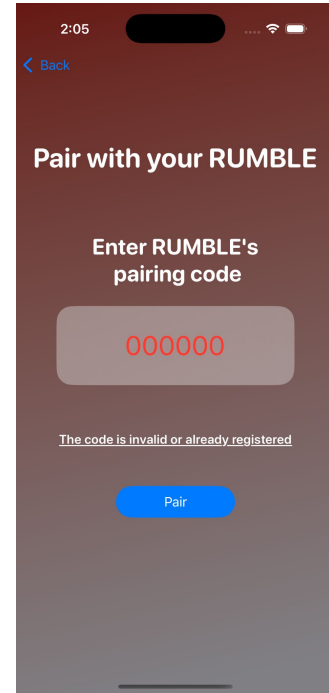
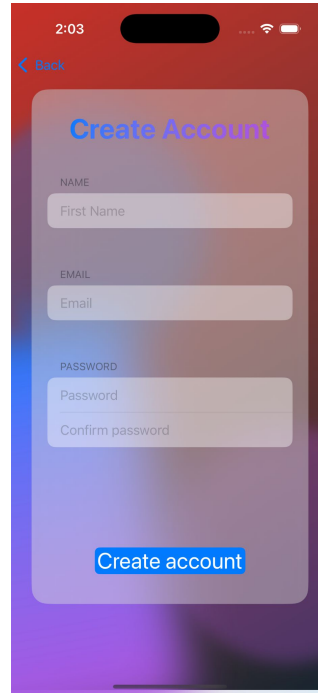
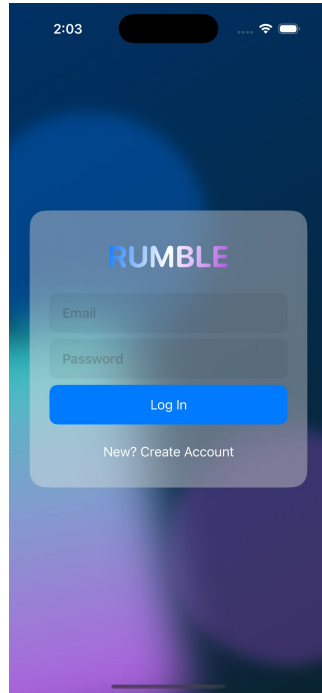
MACHINE LEARNING



BACKEND

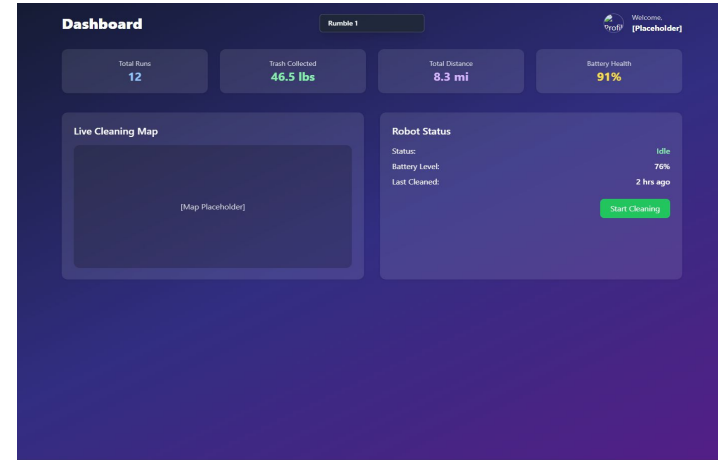
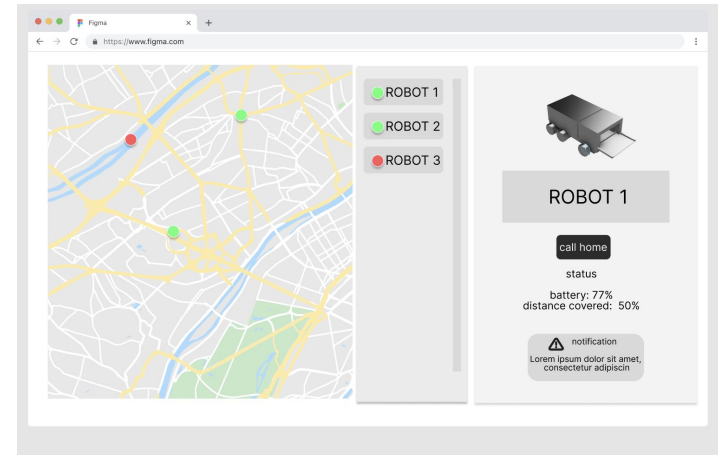


FRONTEND



FRONTEND

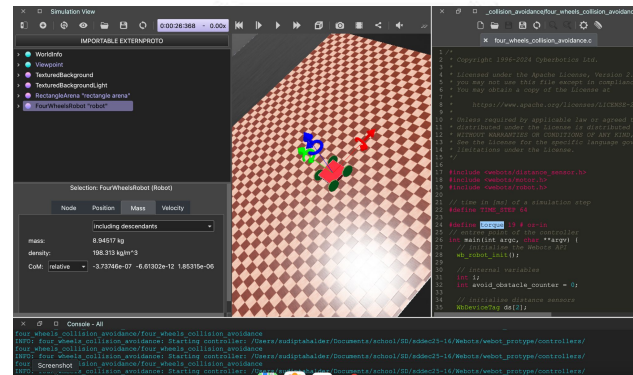
- Followed human-centered design principles.
- Used Figma for early dashboard layout and UI planning
- Used basic HTML, CSS, and Javascript.



Testing

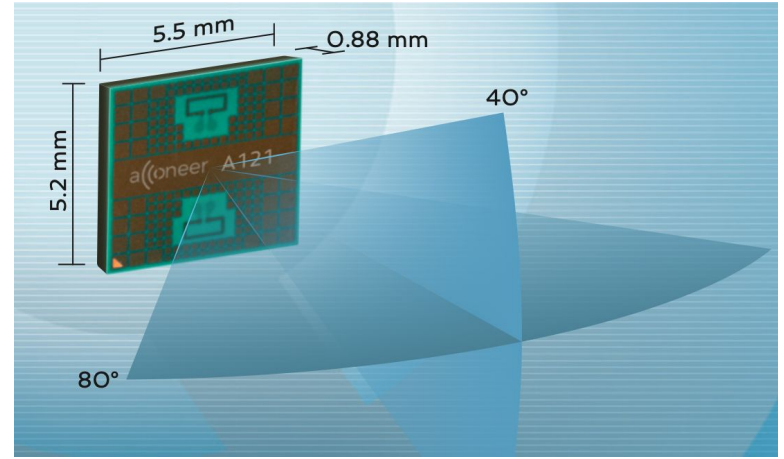
SOFTWARE TESTING

- Unit testing
- Interface Testing
- Integration testing
- Software Input Output
- System Testing
- Machine Learning Benchmark



HARDWARE TESTING

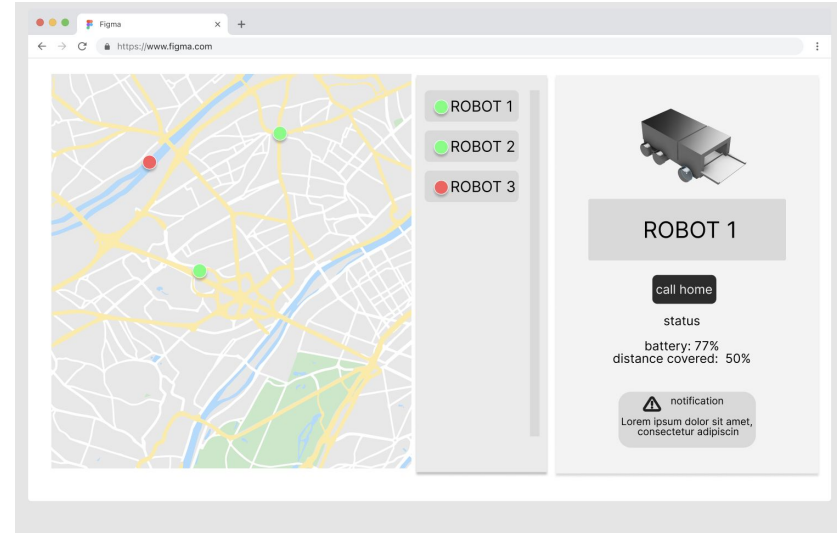
- Component Testing
 - Motors
 - Sensors
 - GPS
- Integration Testing
- Power Management
 - Capacity X Power Usage
 - Environment X Power Usage



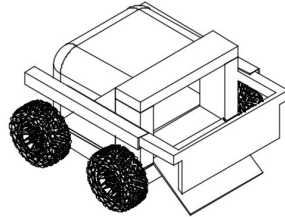
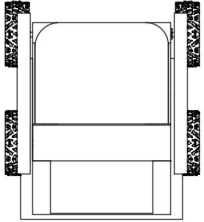
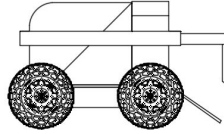
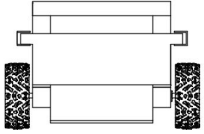
PROTOTYPE EVALUATION

What we are looking for:

- Usability
- Average Clean Time
- Battery Life
- Sensor Accuracy



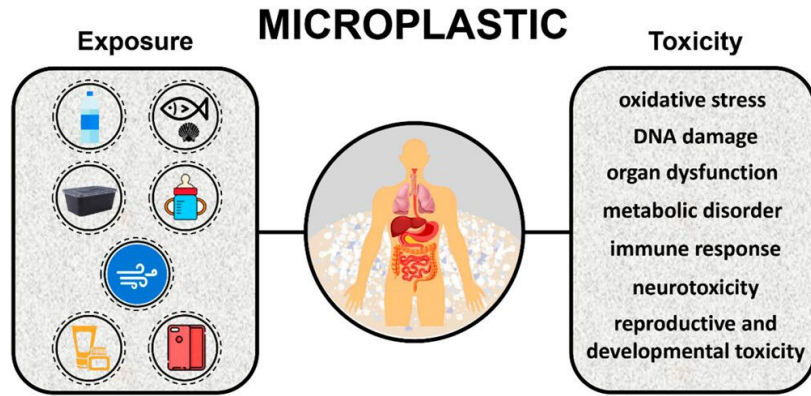
System Testing



- Customer Survey
 - Battery Capacity
 - Pickup Mechanism
 - Sensor Accuracy

QUESTIONS ?

Reference Images



[Photo Link](#)



[Photo Link](#)

Reference Images



[Photo Link](#)

IP Rating Chart

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 3 Protected against a solid object greater than 2.5mm, such as a wire	 3 Protected against water spray at 60 degree angle
 4 Protected against a solid object greater than 1.0mm, such as a thin strap	 4 Protected against water splashing from any angle
 5 Dust Protected. Prevents ingress of dust sufficient to cause harm	 5 Protected against water and from any angle
 6 Dust tight. No ingress of dust	 6 Protected against powerful water jets and heavy seas
Example:	
 + 	 7 Protected against the effects of temporary submersion in water. (30 minutes at 1 meter)
IP 65	 8 Protected against the effects of permanent submersion in water. (up to 15 feet)

[Photo Link](#)



Selenium

[Photo Link](#)

Reference Images

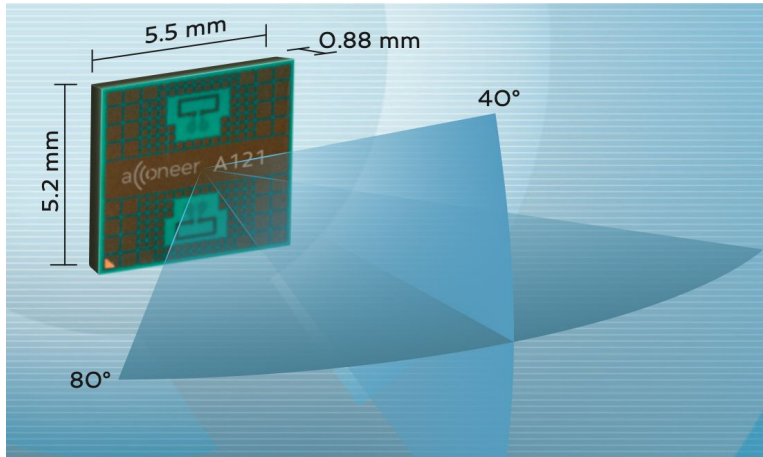


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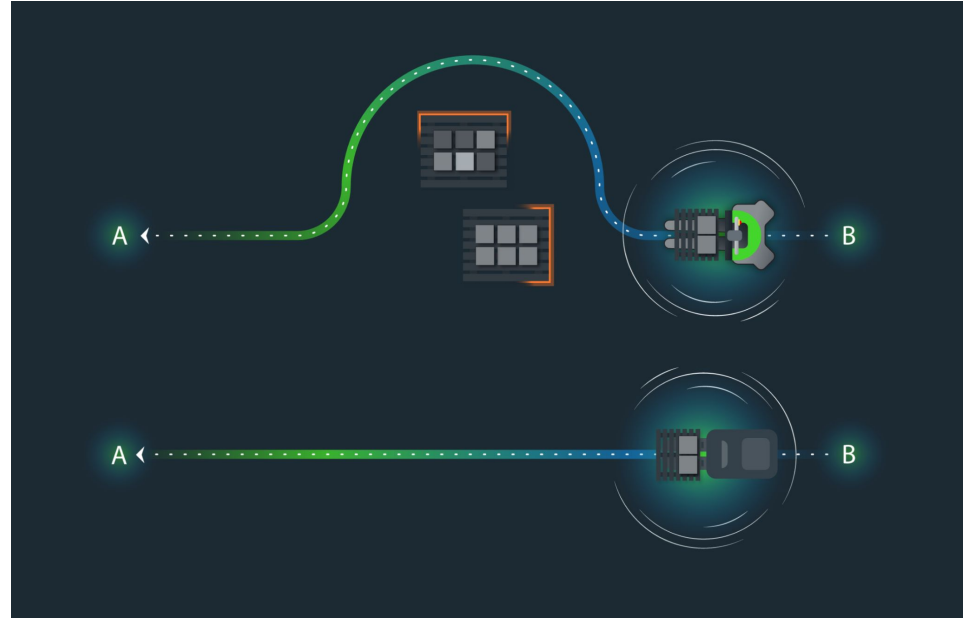


Photo Link

Reference Images

<https://thenounproject.com/icon/litter-1339736/>

<https://www.theguardian.com/environment/2025/feb/11/microplastics-mice-brains>

<https://www.vecteezy.com/free-vector/river-logo>

<https://www.vecteezy.com/vector-art/2285734-river-mountain-landscape-logo-design>

